

Use angles on a line and around a point

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Angles around a point are 85 degrees, 140 degrees and x . Find x . Copy or complete the first step.

2. One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.

3. A triangle has angles 47 degrees and 68 degrees. Find the third angle.

4. Miro says: Miro estimates from the drawing even when the diagram is not to scale. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Angles around a point are 85 degrees, 140 degrees and x . Find x . Copy or complete the first step.
Answer 135 degrees.
2. One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.
Answer 68 degrees.
3. A triangle has angles 47 degrees and 68 degrees. Find the third angle.
Answer 65 degrees.
4. Miro says: Miro estimates from the drawing even when the diagram is not to scale. Is this correct? Explain.
Answer Use labelled values and known geometric facts, then check the total angle relationship.

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Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Angles around a point are 85 degrees, 140 degrees and x . Find x .

6. Check this result using an inverse, estimate or known fact: A triangle has angles 47 degrees and 68 degrees. Find the third angle.

2. One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.

7. Prove or explain your reasoning: A circle has radius 4.5 cm. What is its diameter?

3. A triangle has angles 47 degrees and 68 degrees. Find the third angle.

8. Identify and correct this misconception: Miro estimates from the drawing even when the diagram is not to scale.

4. Solve this independently and show every stage: Angles around a point are 85 degrees, 140 degrees and x . Find x .

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.

10. Write and solve a similar question to: A triangle has angles 47 degrees and 68 degrees. Find the third angle.

Teacher answers and checking prompts

1. Angles around a point are 85 degrees, 140 degrees and x . Find x .
Answer 135 degrees.
2. One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.
Answer 68 degrees.
3. A triangle has angles 47 degrees and 68 degrees. Find the third angle.
Answer 65 degrees.
4. Solve this independently and show every stage: Angles around a point are 85 degrees, 140 degrees and x . Find x .
Answer 135 degrees.
5. Use a second method or representation for: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.
Answer Expected result: 68 degrees. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A triangle has angles 47 degrees and 68 degrees. Find the third angle.
Answer Expected result: 65 degrees. A valid check must be shown.
7. Prove or explain your reasoning: A circle has radius 4.5 cm. What is its diameter?
Answer 9 cm. The explanation must show why the method works.
8. Identify and correct this misconception: Miro estimates from the drawing even when the diagram is not to scale.
Answer Use labelled values and known geometric facts, then check the total angle relationship.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A triangle has angles 47 degrees and 68 degrees. Find the third angle.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

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Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: A circle has radius 4.5 cm. What is its diameter?

6. Create a new example that tests the same idea as: A triangle has angles 47 degrees and 68 degrees. Find the third angle.

2. Prove the answer to this question, rather than only calculating it: Angles around a point are 85 degrees, 140 degrees and x . Find x .

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 65 degrees.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro estimates from the drawing even when the diagram is not to scale.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: A circle has radius 4.5 cm. What is its diameter?
Answer 9 cm. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Angles around a point are 85 degrees, 140 degrees and x . Find x .
Answer Expected result: 135 degrees. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.
Answer Expected result: 68 degrees. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 65 degrees.
Answer One valid reconstruction is: A triangle has angles 47 degrees and 68 degrees. Find the third angle. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro estimates from the drawing even when the diagram is not to scale.
Answer Use labelled values and known geometric facts, then check the total angle relationship.
6. Create a new example that tests the same idea as: A triangle has angles 47 degrees and 68 degrees. Find the third angle.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.